UE26MA141A Engineering Mathematics – I · Unit 2 · Higher-Order Linear Differential Equations

Graphs and diagrams

Every graph and diagram in scope, redrawn as clean vector figures.

Class 001 · Introduction to second and higher order DE with constant coefficients · compiled 15 October 2026

ESA ★★★	Asked repeatedly in past ESA papers
ESA ★★	Asked in past ESA papers or computed in class-work problems
ESA ★	Conceptual: taught in class, not examined so far
added for clarity	Standard context that is not in the class material
transcript only	Said in the lecture but not on any slide
video only	Seen only in a frame of the class video
not on current slides	From older material or past papers, not this year's slides
S001·12	Unit 2 deck 001, slide 12 (rendered .pptx)
T001 00:17:38	Unit 2 AV-summary transcript 001 (“Introduction to higher order differential equations, complementary function and particular integral”), segment start
ESA Dec-23 3a	Past paper: UE22MA141A Dec 2023, question 3.a
ESA Jan-24 s3 3d	Past paper: UE23MA141A Jan 2024 set 3, question 3.d

Contents

1. Figure G4.1 · Superposition: a combination of solutions is a solution
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3. Figure G5.1 · Every solution is CF + PI: one PI, many CFs

Colours follow the Okabe–Ito order: blue = first solution, vermilion (dashed) = second, green = third or the combination; grey dashed lines are references. The deck's two classification trees (slides 001·5 and 001·10) are redrawn as flowcharts F1.1 and F2.1. The deck has no graphs; the three graphs are added for clarity and every plotted solution was checked with sympy.

4 Linear independence and the complementary function

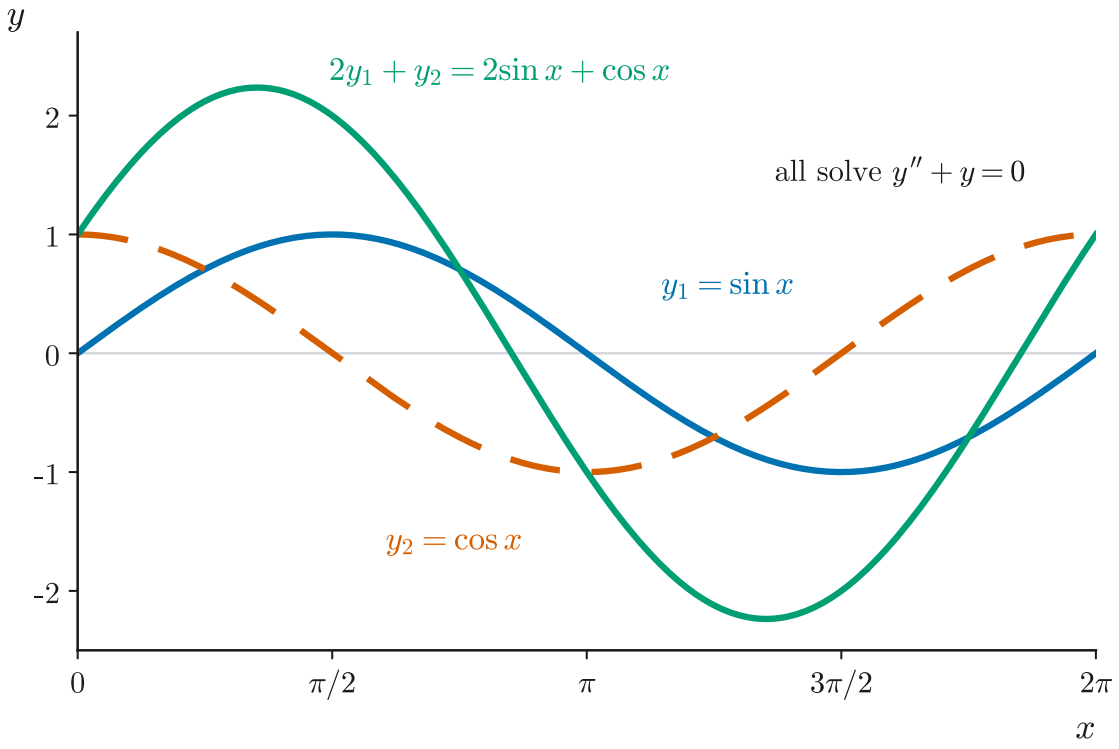


Figure G4.1. Superposition: a combination of solutions is a solution. ESA ★★★

ADDED FOR CLARITY S001·15 · S001·16

How to read it. For $y'' + y = 0$, $y_1 = \sin x$ (blue) and $y_2 = \cos x$ (vermilion dashed) are independent solutions. Their combination $2\sin x + \cos x$ (green) is again a solution; it has a different amplitude and phase, so it is not a multiple of either. Varying c_1, c_2 gives every solution: the CF $c_1 \sin x + c_2 \cos x$ (sympy dsolve).

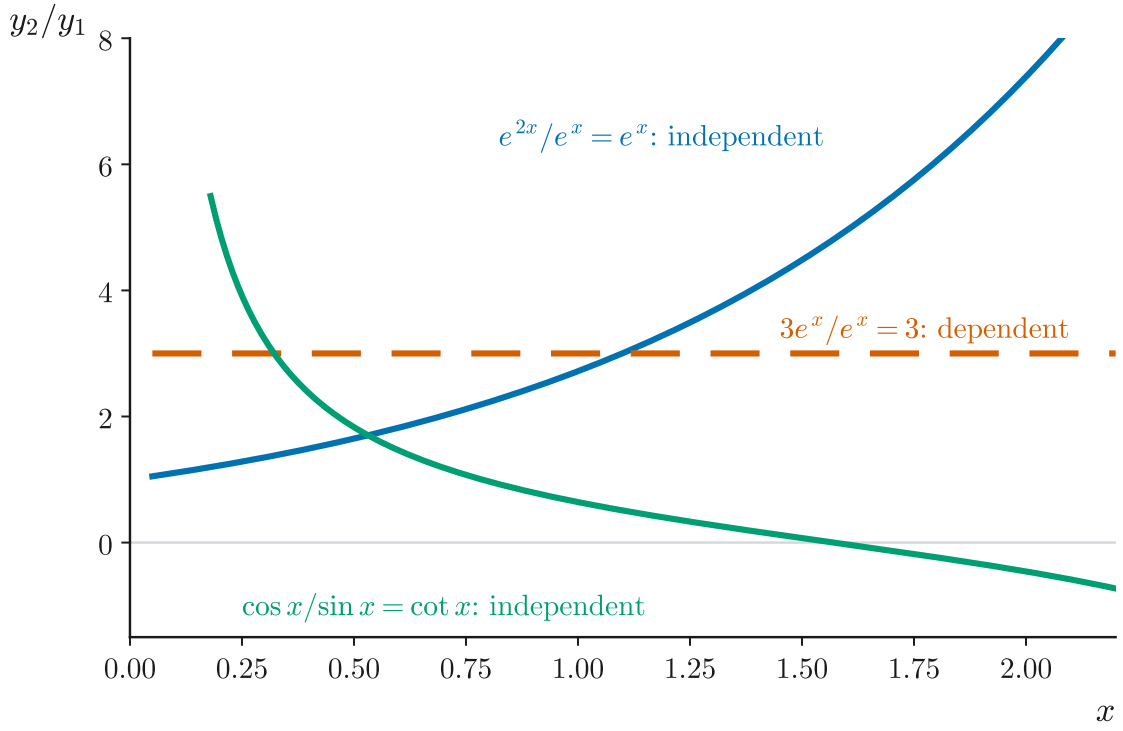


Figure G4.2. The ratio test for two functions. ESA ★★

ADDED FOR CLARITY S001·14 · T001 00:22:00

How to read it. Plot y_2/y_1. For e^x and $3e^x$ the ratio is the constant 3 (vermilion dashed): dependent. For e^x and e^{2x} it is e^x (blue), and for $\sin x$ and $\cos x$ it is $\cot x$ (green): not constant, so independent. A constant ratio is exactly linear dependence.

5 The general solution: $y = \text{CF} + \text{PI}$

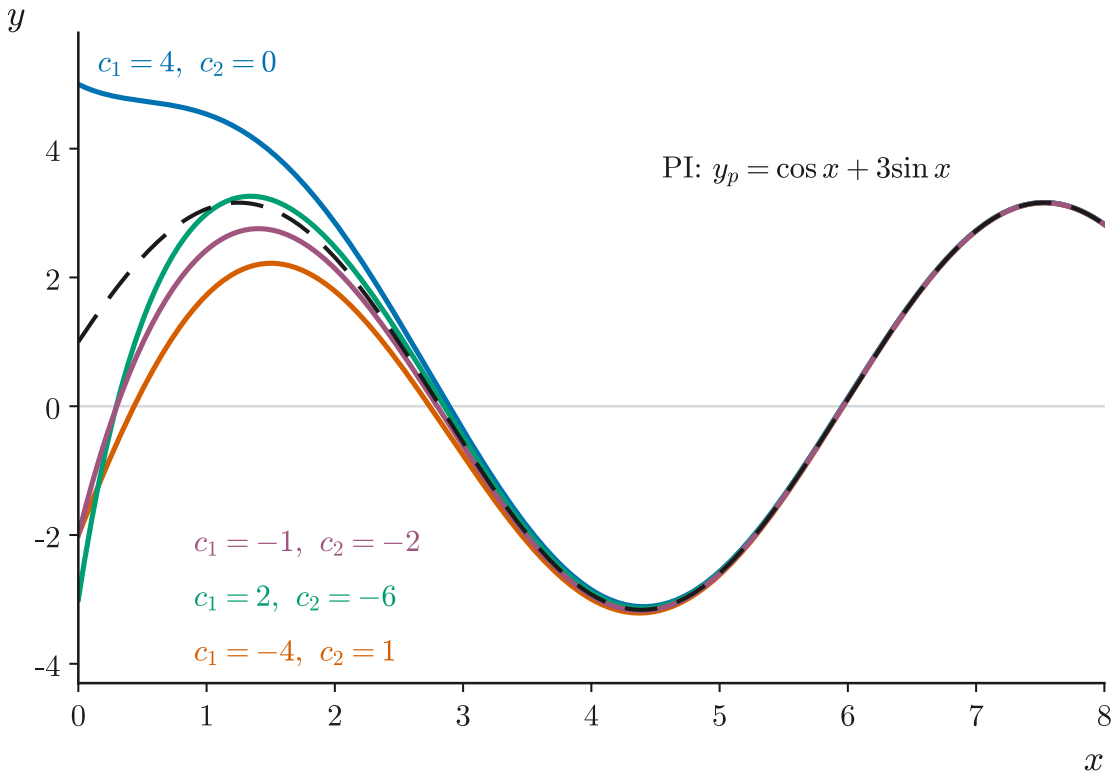


Figure G5.1. Every solution is CF + PI: one PI, many CFs. ESA ★★★

ADDED FOR CLARITY S001·17 · S001·12

How to read it. For $y'' + 3y' + 2y = 10\cos x$ the PI is $y_p = \cos x + 3\sin x$ (black dashed) and the CF is $c_1 e^{-x} + c_2 e^{-2x}$. Four choices of c_1, c_2 (coloured curves) start at different values but all approach y_p, because here the CF decays. The constants come from the CF only; the PI is the same for all. Every curve was checked with sympy dsolve.